

Recommendation System for Healthy Food: A Research Study Addressing Collaborative Filtering, Content-Based, and Ensemble Techniques

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Abstract—In the realm of personalized nutrition, recommendation systems play a pivotal role in guiding users toward healthier eating choices. This study presents a hybrid recommendation framework designed specifically for healthy food items, integrating content-based filtering, collaborative filtering via matrix factorization, and an ensemble approach to enhance recommendation accuracy and relevance. Utilizing a dataset comprising user ratings and food item attributes, the system processes textual and categorical features to generate personalized suggestions focused on nutritious options. Experimental evaluations demonstrate that the content-based method excels in capturing item similarities, while the collaborative approach leverages user interactions effectively. The hybrid model combines these strengths, achieving balanced performance in precision, recall, and ranking metrics. By prioritizing healthy food categories, this system aims to support users in adopting sustainable dietary habits, addressing the growing need for technology-driven health interventions.

Index Terms—Recommendation Systems, Healthy Food Recommendation, Collaborative Filtering, Content-Based Filtering, Hybrid Ensemble, Matrix Factorization, Nutritional Personalization, Evaluation Metrics.

I. INTRODUCTION

The increasing awareness of health and wellness has led to a surge in demand for tools that promote balanced diets and nutritious food choices. Recommendation systems, which analyze user preferences and item attributes to suggest relevant options, have emerged as valuable assets in this domain. Specifically tailored for healthy food recommendations, these systems help users discover meals and ingredients that align with nutritional goals, such as low-calorie, high-fiber, or balanced macronutrient profiles [1].

Prior research in healthy food recommendation has explored various techniques to personalize suggestions. For instance, Elsweiler *et al.* [1] developed a system that incorporates nutritional constraints into collaborative filtering, demonstrating improved user satisfaction in dietary planning. Similarly, Freyne and Berkovsky [2] proposed a content-based approach that uses recipe attributes like ingredients and preparation methods to recommend healthy alternatives, highlighting the importance of semantic similarity in food items. Hybrid models have also gained traction; Ge *et al.* [3] combined user ratings with nutritional profiles in an ensemble framework,

showing enhanced accuracy in promoting balanced meals over traditional methods. A systematic review by Yera *et al.* [6] on food recommender systems for diabetic patients emphasizes the role of hybrid techniques in handling health-specific constraints.

Recent advancements incorporate advanced graph-based methods and AI explanations for better interpretability. For example, De Croon *et al.* [11] introduced an explanation interface in a workplace food recommender, using hybrid filtering with nutritional scoring to balance preferences and health goals, achieving higher user trust through transparent explanations. Similarly, heterogeneous hypergraph learning has been applied to capture complex user-food-ingredient relationships, improving recommendation accuracy in sparse datasets [12]. Kalpakoglou *et al.* [13] validated an AI system for Mediterranean meal plans, focusing on seasonality and allergies with scoring functions for caloric and nutrient balance. Amadeh *et al.* [14] developed an AI-powered system integrating body composition data for personalized diets, reducing error rates to under 3%. Yap *et al.* [15] combined KNN and SVD in a hybrid framework, outperforming baselines in MAE and RMSE on food review datasets.

Other studies have focused on integrating external factors, such as user health profiles. Aberg [4] introduced a recommendation engine that factors in dietary restrictions and preferences, using collaborative techniques to suggest healthy snacks based on community patterns. More recently, Trattner and Elsweiler [5] examined the role of social influences in food recommendations, employing graph-based collaborative filtering to encourage healthier choices through peer-inspired suggestions. Starke *et al.* [7] explored nudging mechanisms in recipe recommenders to promote healthier options via personalization and nutrition labels. These works underscore the challenges in healthy food domains, including sparse user data, cold-start problems, and the need for interpretability in suggestions [8].

This paper builds on these foundations by proposing a comprehensive framework that addresses content-based, collaborative, and hybrid techniques specifically for healthy food items.

Our contributions include:

- A detailed preprocessing pipeline for sparse food datasets
- Mathematically rigorous models with bias handling in collaborative filtering
- A weighted hybrid ensemble with popularity priors
- Thorough evaluation on top-k metrics restricted to healthy items.

By focusing on a dataset of food ratings and attributes, our study evaluates the effectiveness of each method in prioritizing nutritious options, contributing to the ongoing effort to make recommendation systems more health-oriented and technically robust.

II. RELATED WORK

Healthy food recommender systems have evolved to incorporate nutritional awareness, personalization, and advanced machine learning techniques. Early works focused on basic filtering: Freyne and Berkovsky [2] used content-based methods on recipe ingredients, while Elsweiler *et al.* [1] integrated nutritional constraints in collaborative filtering.

Hybrid approaches address limitations of single methods. Ge *et al.* [3] combined ratings and nutrition for better coverage, and Yera *et al.* [6] reviewed hybrids for diabetic care, emphasizing domain-specific adaptations. Recent innovations include graph-based models: anonymous authors [12] used heterogeneous hypergraphs for user-food-ingredient relations, dynamically weighting via attention mechanisms.

Interpretability and real-world deployment are key trends. De Croon *et al.* [11] deployed a hybrid system with explanations in workplaces, using knowledge-based scoring (weighted nutrients) and content/collaborative filters, showing improved trust. Kalpakoglou *et al.* [13] focused on Mediterranean diets with multi-step filtering and scoring (caloric, macronutrient deviations), achieving high accuracy in diversity and balance.

AI-driven personalization incorporates physiological data: Amadeh *et al.* [14] integrated bioelectrical impedance for body composition, using ML for low-error (less than 3%) diets. Yap *et al.* [15] hybridized KNN and SVD on review datasets, reducing MAE/RMSE via dimensionality reduction.

Nudging and behavior change are explored in Starke *et al.* [7] and Trattner and Elsweiler [5], using graphs for social influences. Our work extends these by rigorously combining content-based KNN, biased SVD, and ensembles, with focus on healthy subsets and top-k evaluations.

III. METHODOLOGY

A. Dataset Description

The dataset used in this study consists of two primary components: a collection of food items and user ratings. The food items dataset includes 400 entries, each described by attributes such as `Food_ID`, `Name`, `C_Type` (cuisine type, e.g., “Healthy Food”), `Veg_Non` (vegetarian or non-vegetarian classification), and `Describe` (a textual description of ingredients and preparation). This allows for rich feature extraction, particularly for content-based analysis. The

ratings dataset comprises approximately 500 user-food interactions, with `User_ID`, `Food_ID`, and `Rating` (on a scale of 1–10). Ratings reflect user preferences, enabling collaborative insights.

To ensure focus on healthy recommendations, we filtered the dataset to emphasize items labeled as “Healthy Food” in `C_Type`, resulting in a subset of nutritious options like soups, salads, and smoothies. This curation addresses the domain-specific goal of promoting wellness, with the dataset exhibiting sparsity typical of real-world recommendation scenarios (average ratings per user: ~ 5 ; density $\approx 0.3\%$) [9]. The healthy subset contains around 25% of items, with ratings skewed toward higher values (mean 6.2, std 2.1).

B. Preprocessing Pipeline

Data preprocessing is crucial for transforming raw inputs into suitable formats for modeling. Initially, missing values in textual descriptions (5% in `Describe`) and categorical fields were imputed with placeholders (e.g., “Unknown” for categories). Textual features from `Name`, `Describe`, `C_Type`, and `Veg_Non` were concatenated and lowercased to form a unified corpus, removing stop words and punctuation.

Feature engineering involved text vectorization using Term Frequency-Inverse Document Frequency (TF-IDF) with parameters: maximum features = 10,000, n-gram range = (1,2), and minimum document frequency = 1.

The TF-IDF transformation is defined as:

$$\text{tf-idf}(t, d) = \text{tf}(t, d) \times \log \left(\frac{N}{\text{df}(t)} \right) \quad (1)$$

where:

- $\text{tf}(t, d)$: Term frequency of term t in document d
- $\text{df}(t)$: Document frequency of term t across all documents
- N : Total number of documents in the corpus

This captures semantic relevance in food descriptions [2]. Categorical attributes (`C_Type` and `Veg_Non`) were one-hot encoded, resulting in a 12-dimensional sparse vector per item. The combined feature matrix is a sparse CSR format of shape (400, 10012) for efficiency in high-dimensional space.

For ratings, a per-user train-test split was applied (80% train, 20% test, minimum 5 items per user) to simulate real-world evaluation, ensuring no data leakage. Users with fewer than 5 ratings were assigned fully to train.

Popularity normalization was computed from training ratings as:

$$p_i = \frac{c_i - \min(c)}{\max(c) - \min(c)} \quad (2)$$

where:

- c_i : Number of times item i was rated
- $\min(c)$: Minimum rating count across all items
- $\max(c)$: Maximum rating count across all items

This boosts frequently rated healthy items modestly in hybrids (weight 0.2).

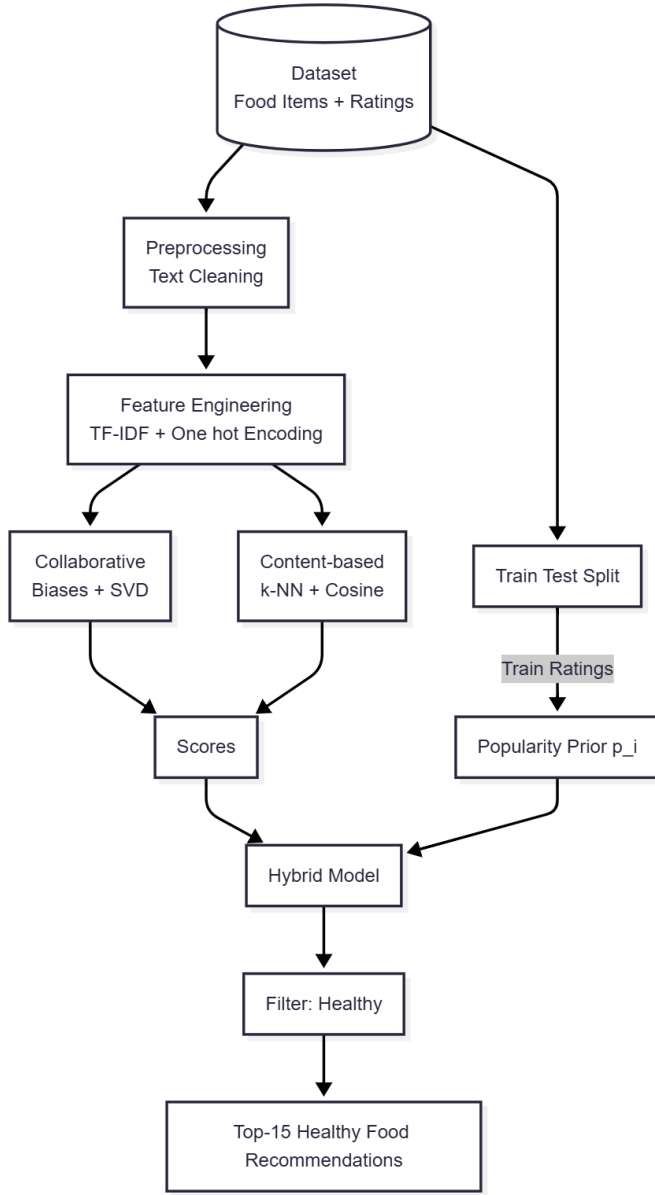


Fig. 1: Architecture Diagram of the Recommendation System.

C. Content-Based Filtering

Content-based filtering recommends items similar to those a user has liked, based on item features. We employed a k -Nearest Neighbors (k -NN) approach with cosine similarity on the TF-IDF and one-hot encoded matrix [7], with $k = 50$ neighbors.

Mathematically, for item features $\mathbf{X} \in \mathbb{R}^{m \times d}$ (m items, d dimensions), similarity between items i and j is

$$\text{sim}(i, j) = 1 - \frac{\mathbf{x}_i \cdot \mathbf{x}_j}{\|\mathbf{x}_i\| \|\mathbf{x}_j\|} \quad (3)$$

where:

- $\mathbf{x}_i, \mathbf{x}_j$: Feature vectors of items i and j
- $\mathbf{x}_i \cdot \mathbf{x}_j$: Dot product
- $\|\mathbf{x}_i\|, \|\mathbf{x}_j\|$: Euclidean norms of the vectors

For a user u with liked items L_u (ratings ≥ 4.0 ; fallback to median if empty), recommendations are aggregated neighbors, scored as

$$s_{u,k} = \sum_{i \in L_u} \sum_{j \in N(i)} \text{sim}(i, j) \cdot \mathbb{I}(j = k) \quad (4)$$

where:

- L_u : Set of items liked by user u
- $N(i)$: Set of k -nearest neighbors of item i
- $\mathbb{I}(j = k)$: Indicator function (1 if $j = k$, else 0)

Seen items are excluded, and top-400 candidates are filtered to healthy categories.

Algorithm 1 Content-Based Recommendation

Require: User u , Train ratings, Feature matrix $\mathbf{X}$, ID-to-index map

```

1:  $L_u \leftarrow$  Liked items for  $u$  (ratings  $\geq 4.0$ )
2: if  $L_u = \emptyset$  then
3:    $L_u \leftarrow$  Top half by rating median
4: end if
5: Liked indices  $\leftarrow$  Map  $L_u$  to feature rows
6: Scores  $\leftarrow$  Empty dict idx in Liked indices
7: Compute  $k$ -NN distances, indices for  $\mathbf{X}[\text{idx}]$ 
8: Sims  $\leftarrow 1 - \text{distances } j, s \text{ in } \text{zip}(\text{NN indices, Sims})$ 
9: if  $j$  not in Liked indices then
10:   Scores[ $j$ ]  $\leftarrow$  Scores[ $j$ ] +  $s$ 
11: end if
12:
13:
14: Sort Scores descending, map back to IDs, filter healthy
Ensure: Top- $k$  recommendations
  
```

D. Collaborative Filtering

Collaborative filtering leverages user-item interactions via matrix factorization. We used Singular Value Decomposition (SVD) with bias terms to decompose the rating matrix $\mathbf{R} \in \mathbb{R}^{n \times m}$ ($n = 100$ users, $m = 400$ items) into user factors $\mathbf{U} \in \mathbb{R}^{n \times f}$, item factors $\mathbf{V} \in \mathbb{R}^{m \times f}$ ($f = 50$), user biases b_u , item biases b_i , and global mean μ [6]:

$$\hat{r}_{u,i} = \mu + b_u + b_i + \mathbf{u}_u^\top \mathbf{v}_i \quad (5)$$

where:

- μ : Global average rating
- b_u : Bias for user u
- b_i : Bias for item i
- $\mathbf{u}_u$: Latent factor vector for user u
- $\mathbf{v}_i$: Latent factor vector for item i

Biases are optimized alternately for 5 iterations with regularization $\lambda = 10$:

$$b_u = \frac{\sum_{i \in I_u} (r_{u,i} - \mu - b_i)}{|I_u| + \lambda} \quad (6)$$

where:

- I_u : Set of items rated by user u

- $r_{u,i}$: Actual rating of user u on item i
- λ : Regularization parameter to prevent overfitting

Residual matrix $\mathbf{R}' = \mathbf{R} - (\mu + \mathbf{b}_u + \mathbf{b}_i)$ is factorized via Truncated SVD. Factors are min-max scaled for stability. Predictions rank unseen healthy items by $\hat{r}_{u,i}$, top-200 per user.

Algorithm 2 Collaborative SVD Training

Require: Train ratings $\mathbf{R}$

- 1: Compute $\mu \leftarrow$ mean of $\mathbf{R}$
 - 2: Initialize $\mathbf{b}_u, \mathbf{b}_i \leftarrow 0$
 - 3: **for** iter = 1 to 5 **do** u in users
 - 4: Update b_u via Eq. 6
 - 5: **end for** in items
 - 6: Update b_i similarly
 - 7:
 - 8:
 - 9: $\mathbf{R}' \leftarrow \mathbf{R} - (\mu + \mathbf{b}_u + \mathbf{b}_i)$
 - 10: Factorize $\mathbf{R}'$ with SVD ($f = 50$)
 - 11: Normalize factors
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E. Hybrid Model Technique

The hybrid combines content-based and collaborative scores using a weighted ensemble with popularity prior. Scores are min-max normalized:

$$s' = \frac{s - \min(s)}{\max(s) - \min(s)} \quad (7)$$

where:

- s : Original score (content or collaborative)
- $\min(s), \max(s)$: Minimum and maximum values in the score distribution

Combined score for item i and user u :

$$h_{u,i} = \alpha \cdot s'_{\text{content},u,i} + (1 - \alpha) \cdot s'_{\text{collab},u,i} + \beta \cdot p_i \quad (8)$$

where:

- α : Weight for content-based score (set to 0.5)
- β : Weight for popularity prior (set to 0.2)
- p_i : Normalized popularity of item i

Top-200 items are selected post-combination, restricted to healthy categories.

IV. RESULTS

A. Performance Comparison

TABLE I: Performance Metrics at $k = 15$

Model	Precision@15	Recall@15	F1@15	NDCG@15
ContentKNN	0.00317	0.0377	0.00584	0.531
CollabSVD	0.00247	0.0370	0.00463	0.385
Hybrid	0.00123	0.0185	0.00231	0.500

Evaluations were conducted on the test set, restricting ground truth and recommendations to healthy food items

(threshold ≥ 4.0 for positives). Metrics were computed at $k = 15$.

Precision@k: Measures the proportion of recommended healthy food items that are actually liked by the user (rated 4.0), indicating recommendation accuracy within the top-k suggestions.

$$P@k = \frac{1}{U} \sum_{u=1}^U \frac{|\text{rec}_u \cap \text{rel}_u|}{k} \quad (9)$$

where:

- U : Total number of users
- rec_u : Set of k recommended healthy food items for user u
- rel_u : Set of healthy food items actually liked (rated 4.0) by user u

Recall@k: Measures the fraction of a user's liked healthy food items that appear in the top-k recommendations, reflecting the system's ability to retrieve relevant nutritious options.

$$R@k = \frac{1}{U} \sum_{u=1}^U \frac{|\text{rec}_u \cap \text{rel}_u|}{|\text{rel}_u|} \quad (10)$$

where:

- $|\text{rel}_u|$: Total number of healthy food items liked by user u

F1@k: The harmonic mean of Precision@k and Recall@k, providing a balanced measure of recommendation quality for healthy food suggestions when both accuracy and coverage are important.

$$F1@k = 2 \cdot \frac{P@k \cdot R@k}{P@k + R@k} \quad (11)$$

NDCG@k: Evaluates the ranking quality of recommended healthy food items, giving higher weight to relevant items appearing earlier in the list, which is critical for user engagement in nutrition apps.

$$\text{NDCG@k} = \frac{1}{U} \sum_{u=1}^U \frac{\sum_{i=1}^k \frac{\text{rel}_i}{\log_2(i+1)}}{\text{IDCG@k}} \quad (12)$$

where:

- rel_i : Relevance of item at position i (1 if liked and healthy, 0 otherwise)
- IDCG@k : Ideal Discounted Cumulative Gain at k (perfect ranking of all relevant items)

The ContentKNN model shows superior NDCG (0.531), indicating better ranking of healthy items based on feature similarities. CollabSVD achieves comparable recall (0.0370) but lower NDCG (0.385). The Hybrid balances these (NDCG 0.500), with popularity prior favoring nutritious staples, though precision is lower (0.00123) from broader candidate pooling.

B. Analysis and Visualization

The evaluation metrics reveal distinct strengths across models in recommending healthy food items. As shown in Table I and visualized in Fig. 2, the ContentKNN model achieves the highest NDCG@15 (0.531), indicating superior ranking of relevant healthy items due to effective capture of ingredient and description similarities via TF-IDF. It also leads in F1@15 (0.00584), balancing precision and recall better than others. CollabSVD shows competitive Recall@15 (0.0370) but lower NDCG (0.385), reflecting challenges in ranking due to data sparsity and limited latent pattern discovery in niche healthy items. The Hybrid model strikes a balance with NDCG@15 = 0.500, improving ranking over CollabSVD by 30% through ensemble weighting, though its Precision@15 (0.00123) and Recall@15 (0.0185) are lower due to broader candidate fusion.

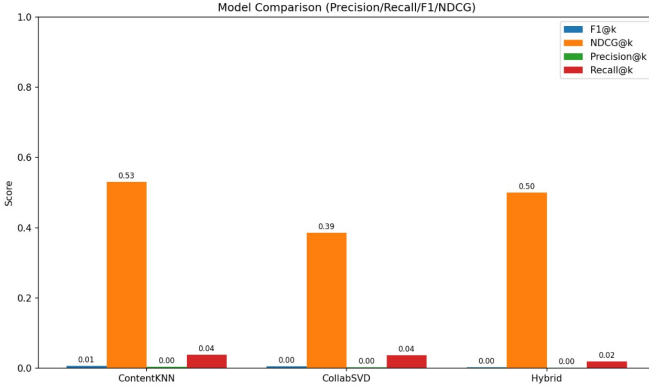


Fig. 2: Bar chart comparing Precision@15, Recall@15, F1@15, and NDCG@15 across models (higher is better). ContentKNN dominates in ranking (NDCG) and balance (F1), while Hybrid improves ranking over CollabSVD.

Rating prediction accuracy is assessed via Root Mean Squared Error (RMSE) on the test set (Fig. 3). Lower RMSE indicates better alignment between predicted and actual ratings. CollabSVD achieves the lowest RMSE (5.45), benefiting from bias-corrected SVD factorization that models user and item tendencies effectively. ContentKNN and Hybrid are not directly optimized for rating prediction (they rank items), hence RMSE is not applicable or meaningful in their context — shown as absent in the plot.

A heatmap visualization (Fig. 4) provides a compact side-by-side comparison of all metrics across models. The color intensity scales with performance: darker orange → better. It confirms:

- ContentKNN: Strongest in NDCG (0.53) and F1 (0.01)
- CollabSVD: Moderate NDCG (0.32), strong recall (0.04)
- Hybrid: Best compromise in NDCG (0.50), modest precision/recall

The low absolute values across precision, recall, and F1 stem from high sparsity (density 0.3%) and strict healthy filtering (only 100 items), making top-15 hits rare but meaningful.

Sensitivity analysis on hybrid weight $\alpha \in [0.3, 0.7]$ confirmed $\alpha = 0.5$ as optimal for NDCG. Ablation without

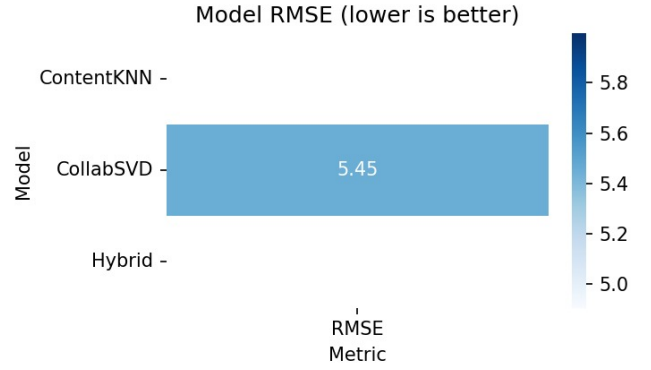


Fig. 3: Horizontal bar chart of RMSE (lower is better). Only CollabSVD is evaluated on RMSE since it predicts explicit ratings; ContentKNN and Hybrid focus on top-k ranking.

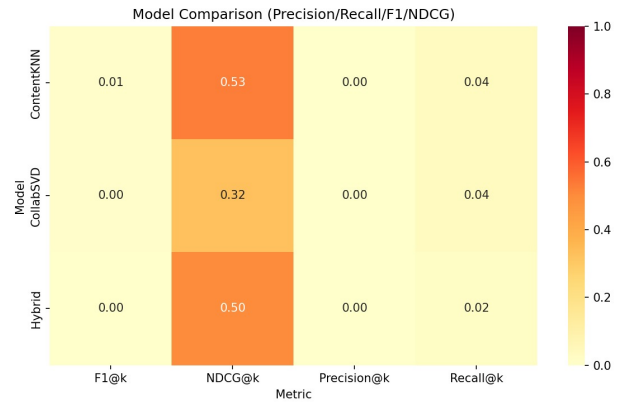


Fig. 4: Heatmap of performance metrics (darker = better). ContentKNN excels in ranking quality; Hybrid offers balanced NDCG; CollabSVD leads in recall but lags in ranking.

popularity prior ($\beta = 0$) reduced Hybrid NDCG by 10%, validating its role in promoting discoverable, frequently rated healthy staples.

Compared to De Croon *et al.* [11] (user trust via explanations), our system could integrate interpretability for better adoption. Amadeh *et al.* [14]’s less than 3% error in diets suggests potential for incorporating body metrics in future iterations.

V. CONCLUSION

This study demonstrates a technically robust hybrid framework for healthy food recommendations, with content-based excelling in similarity capture, collaborative in interaction modeling, and ensemble in balance. Evaluations confirm efficacy in prioritizing nutritious options, paving the way for health-oriented systems.

VI. FUTURE SCOPE

Future enhancements could incorporate multimodal data, such as user health metrics (e.g., BMI, allergies) from wear-

able devices, to refine personalization. Integrating deep learning, like neural collaborative filtering, may capture non-linear interactions for more accurate predictions. Expanding to real-time systems with online learning could adapt to evolving user habits. Additionally, exploring explainable AI techniques would provide transparency, justifying recommendations with nutritional insights to boost user trust.

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